

FUAD AHMED

STATEMENT OF PURPOSE & RESEARCH INTERESTS — MASTER OF SCIENCE IN COMPUTER SCIENCE

Target Institution: Mohamed bin Zayed University of Artificial Intelligence (MBZUAI), Abu Dhabi, UAE • **Intake:** Fall 2027
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Legal Name: TUBA, FUAD AHMED

To the Graduate Admissions Committee and Faculty of Computer Science at MBZUAI,

I am writing to formally express my passionate interest in pursuing the **Master of Science in Computer Science** at the **Mohamed bin Zayed University of Artificial Intelligence (MBZUAI)** for the Fall 2027 intake, and to be considered for the prestigious **MBZUAI Graduate Fellowship**. As a software systems engineer with four years of hands-on industry experience building scalable distributed architectures and low-latency microservices, my graduate objective is to conduct advanced research at the vital intersection of **Distributed Systems, Heterogeneous Hardware Acceleration, and Scalable Infrastructure for Artificial Intelligence**. MBZUAI's world-class faculty, state-of-the-art supercomputing infrastructure in Masdar City, and singular dedication to groundbreaking AI research make it the preeminent institution where I aim to dedicate my graduate studies.

1. ACADEMIC TRAJECTORY & COMPUTATIONAL FOUNDATIONS

My foundation in computing began with an intensive four-year Bachelor of Science in Computer Science curriculum at St. Mary's University in Addis Ababa (Major GPA: 3.20/4.00), conducted strictly in English. My undergraduate coursework grounded me in rigorous discrete mathematics, linear algebra, probability, data structures, operating systems, and computer architecture.

For my senior capstone project, I conceptualized and evaluated a high-concurrency, fault-tolerant distributed data synchronization protocol. Operating across networked Linux nodes, I implemented delta-encoded state serialization and lock-free memory ring buffers in Go and Python, ensuring state consistency across disparate nodes while drastically diminishing network payload overhead and CPU thread contention. This project cemented my conviction that future computational breakthroughs will not emerge from brute-force compute, but from algorithmic elegance and hardware-conscious distributed coordination.

2. PROFESSIONAL ENGINEERING EXPERIENCE & SYSTEMS RESEARCH DRIVERS

Over the past four years, I have worked as a Lead Backend and Distributed Systems Software Engineer. I designed asynchronous microservice pipelines using Python (FastAPI/AsyncIO), Go, and Docker to sustain high-concurrency enterprise workloads. In production environments, I architected distributed Redis caching topologies that reduced database read latency by 48% and engineered resilient message-driven pipelines utilizing Kafka and RabbitMQ to absorb volatile traffic spikes.

Working in developing digital ecosystems taught me that computational scalability must operate under severe constraints of bandwidth and memory footprint. As foundational AI models expand into multi-trillion-parameter regimes, the primary bottleneck in machine learning has shifted from pure statistical modeling to distributed systems engineering: communication overhead, GPU-CPU interconnect latency, memory partitioning, and decentralized parameter synchronization. To bridge this critical divide, I must transition from applied industry backend engineering into rigorous academic systems research.

3. WHY MBZUAI REPRESENTS MY IDEAL RESEARCH ENVIRONMENT

MBZUAI is pioneering the future of artificial intelligence with unmatched vision, institutional agility, and computational power. The university offers an extraordinary research ecosystem that aligns perfectly with my graduate objectives:

- **Pioneering Systems for AI:** MBZUAI's dedicated research into Systems for AI, high-performance distributed training frameworks, and efficient inference pipelines directly addresses the architectural challenges I am determined to investigate. I am eager to explore how low-overhead consensus protocols and asynchronous gradient synchronization can mitigate tail-latency across large-scale accelerator clusters.
- **World-Class Supercomputing Facilities:** Access to MBZUAI's cutting-edge NVIDIA DGX clusters and high-speed InfiniBand interconnects provides an empirical environment impossible to replicate elsewhere, enabling experimental benchmarking of extreme-scale deep learning workloads.
- **Global Impact & Collaborative Culture:** MBZUAI's pioneering foundational models (such as Jais and K2) demonstrate the university's commitment to sovereign, inclusive, and multi-lingual AI innovation. Collaborating with leading international faculty in Masdar City will allow me to hone both theoretical depth and deployable systems architectures.

4. CAREER VISION & CONTRIBUTION TO MBZUAI

Upon graduating with my Master of Science in Computer Science from MBZUAI, I intend to contribute as a Distributed AI Systems Architect and researcher, designing low-latency, energy-efficient computing fabrics that support next-generation intelligent services across the Middle East, Africa, and global tech ecosystems. Long-term, I aspire to pioneer distributed AI infrastructure that makes advanced computing accessible to emerging digital economies.

I bring to MBZUAI proven engineering perseverance, production systems experience, and an unyielding commitment to academic excellence. Supported by the MBZUAI Fellowship, I am ready to contribute actively to the university's vibrant research community.

Respectfully submitted,

Fuad Ahmed (TUBA, FUAD AHMED)

BSc in Computer Science, St. Mary's University

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